UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
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Lecture 19

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1. Metric Entropy and Stochastic Process

1 Metric Entropy and Stochastic Process

Focus We would like to control $\mathbb{E}[\sup_{\theta\in\Theta} U_\theta]$, for some stochastic process $\{U_\theta\}_{\theta\in\Theta}$. The goal is to relate the supremum to the **metric properties** of Θ.

1.1 Metric Entropy

Definition 1 (Metric Space)**.** *A* metric space *is a pair (T, d) where T is a set and $d : T \times T \to \mathcal{R}$ satisfies the following:*

(a) (Non-negativity) $d(\theta, \theta') \geq 0$ for all $\theta, \theta' \in T$, with equality if and only if $\theta = \theta'$.

(b) (Symmetry) $d(\theta, \theta') = d(\theta', \theta)$ for all $\theta, \theta' \in T$.

(c) (Triangle inequality) For any $\theta, \theta', \theta'' \in T$,

$$d(\theta, \theta'') \leq d(\theta, \theta') + d(\theta', \theta'')$$

Remark (Pseudometric)**.** *Any function d satisfying (a)–(c) above, except*

$$d(\theta, \theta') = 0 \iff \theta = \theta'$$

*is called a **pseudometric**.*

Example :

(i) (Metric example) For functions $f, g : [0,1] \to \mathbb{R}$,

$$d(f,g) = \|f-g\|_\infty = \sup_{x\in[0,1]} |f(x)-g(x)|$$

(ii) (Pseudometric example)

$$d(f,g) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\big(f(x_i)-g(x_i)\big)^2}$$

Even if $f \neq g$, $d(f,g)$ can be zero if f and g agree on $x_1, \ldots, x_n$.

Definition 2 (Covering Number)**.** *Given a metric space (T,d), an* ε-cover *of T is a set $\{\theta^1, \ldots, \theta^N\} \subset T$ such that for every $\theta \in T$, there exists $i \in \{1, \ldots, N\}$ satisfying*

$$d(\theta, \theta^i) \leq \varepsilon$$

The smallest such N for which an ε-net exists is called the ***covering number*** *of T, denoted as $N(T,d,\varepsilon)$*

Remark.

1. *(Monotonicity) If $\varepsilon \leq \varepsilon'$, then $N(T,d,\varepsilon) \geq N(T,d,\varepsilon')$. In general, $\lim_{\varepsilon\to 0} N(T,d,\varepsilon) = \infty$.*
2. *(Total boundedness) The space T is called* ***totally bounded*** *if*

$$N(T,d,\varepsilon) < \infty, \quad \forall \varepsilon > 0$$

Definition 3 (Packing Number)**.** *Given a metric space (T,d), an* ε-packing *is a set $\{\theta^1, \ldots, \theta^M\} \subset T$ such that for all $i \neq j$,*

$$d(\theta^i, \theta^j) > \varepsilon$$

The largest *M for which an ε-packing of size M exists is called the* ***packing number*** *of T, denoted by $M(T,d,\varepsilon)$.*

Definition 4 (Metric Entropy)**.** *The* ***metric entropy*** *of T is defined as* $\log M(T,d,\varepsilon)$.

The following lemma shows the connections between covering number and packing number.

Lemma 1. *For every $\varepsilon > 0$,*

$$M(T, d, 2\varepsilon) \underbrace{\leq}_{(1)} N(T, d, \varepsilon) \underbrace{\leq}_{(2)} M(T, d, \varepsilon)$$

Proof. For (2), Let $\mathcal{M}$ be a largest ε-packing of T, so $|\mathcal{M}| = M(T, d, \varepsilon)$. Suppose there exists $\theta \in T$ such that $d(\bar{\theta}, \theta) > \varepsilon$ for all $\bar{\theta} \in \mathcal{M}$. Then $\mathcal{M} \cup \{\theta\}$ would also be an ε-packing, contradicting the maximality of $\mathcal{M}$. Hence, every point in T is within ε of some point in $\mathcal{M}$, which means $\mathcal{M}$ forms an ε-net. Therefore,

$$N(T, d, \varepsilon) \leq |\mathcal{M}| = M(T, d, \varepsilon)$$

For (1), Let $\mathcal{M}$ be a 2ε-packing and $\mathcal{N}$ be an ε-net of T. For any $\theta \in \mathcal{M}$, there exists $r \in \mathcal{N}$ such that

$$d(\theta, r) \leq \varepsilon$$

Note that no ε-ball $B(r, \varepsilon)$ can contain distinct $\theta_1, \theta_2 \in \mathcal{M}$, because otherwise

$$d(\theta_1, \theta_2) \leq d(\theta_1, r) + d(r, \theta_2) \leq \varepsilon + \varepsilon = 2\varepsilon,$$

contradicting the definition of a 2ε-packing. Hence, distinct points of $\mathcal{M}$ correspond to distinct elements of $\mathcal{N}$, implying

$$N(T, d, \varepsilon) = |\mathcal{N}| \geq |\mathcal{M}| = M(T, d, 2\varepsilon)$$

□

Example: ℓ_2-ball in $\mathbb{R}^d$ Let $B_2^d = \{v \in \mathbb{R}^d : \|v\|_2 \leq 1\}$. Then from previous lectures, we have proven that the covering number of the ℓ_2-unit ball satisfies

$$N(B_2^d, \|\cdot\|_2, \varepsilon) \leq \left(1 + \frac{2}{\varepsilon}\right)^d$$

Now we claim the lower bound:

$$N(B_2^d, \|\cdot\|_2, \varepsilon) \geq \left(\frac{1}{\varepsilon}\right)^d$$

Proof. Take any ε-cover of size N with centers $\{\theta_1, \ldots, \theta_N\}$, and let $B_i = B(\theta_i, \varepsilon)$. Since the balls cover B_2^d,

$$\sum_{i=1}^{N} \mathrm{Vol}(B_i) \geq \mathrm{Vol}(B_2^d).$$

Each ball has volume $\mathrm{Vol}(B_i) = \varepsilon^d \mathrm{Vol}(B_2^d)$, so

$$N\,\varepsilon^d\,\mathrm{Vol}(B_2^d) \geq \mathrm{Vol}(B_2^d) \quad \Rightarrow \quad N \geq \left(\frac{1}{\varepsilon}\right)^d$$

□

Corollary 1 (Metric Entropy for ℓ_2-ball)**.**

$$\log M(B_2^d, \|\cdot\|_2, \varepsilon) = \Theta\left(d \log \frac{1}{\varepsilon}\right)$$

Example: Binary Hypercube Consider $T = \{0,1\}^d$ equipped with the normalized Hamming distance

$$\bar{d}_H(\theta, \theta') = \frac{1}{d}\sum_{i=1}^{d} \mathbb{1}\{\theta_i \neq \theta_i'\}$$

We aim to bound $N(T, \bar{d}_H, \delta)$ for $\delta < \frac{1}{2}$.
(1) Upper bound: Consider $S \subset T$ where

$$S = \{\theta \in T : \theta_j = 0 \text{ for } j \geq (1-\delta)d\}$$

Clearly, S is a δ-cover. Thus,

$$\log N(T, \bar{d}_H, \delta) \leq \log\left(2^{(1-\delta)d}\right) \quad \Rightarrow \quad \log_2 N(T, \bar{d}_H, \delta) \leq (1-\delta)d.$$

(2) Lower bound: Take any δ-cover $\{\theta^1, \ldots, \theta^N\} \subset T$. It must satisfy

$$\bigcup_{i=1}^{N} B_{\bar{d}_H}(\theta^i, \delta) = T$$

Each Hamming ball has volume

$$|B_{\bar{d}_H}(\theta^i, \delta)| = \sum_{i=1}^{\delta d} \binom{d}{i}$$

Hence,

$$N \cdot \sum_{i=1}^{\delta d} \binom{d}{i} \geq 2^d \Rightarrow N \geq 2^d \frac{1}{\sum_{i=1}^{\delta d}\binom{d}{i}}$$

Since $\sum_{i=1}^{\delta d} \binom{d}{i} = 2^d \Pr\left[\sum_{i=1}^{d} X_i \le \delta d\right]$ for $X_i \sim \text{Bernoulli}(1/2)$, we use the **Chernoff Tail Inequality** to get bound:

$$\Pr\left[\sum_{i=1}^{d} X_i \le \delta d\right] \le e^{-2d(1/2-\delta)^2}$$

Thus, for $\delta < 1/2$,

$$\log N \ge 2d\left(\tfrac{1}{2} - \delta\right)^2$$

1.2 Stochastic Process associated with T

Metric entropy enables us to study $\mathbb{E}[\sup_{\theta \in T} U_\theta]$ for some stochastic process $\{U_\theta\}_{\theta \in T}$.

Definition 5 (Canonical Gaussian Process)**.** *Fix a subset $T \subset \mathbb{R}^d$. The **canonical Gaussian process** associated with T is defined as*

$$\mathcal{G}(T) = \mathbb{E}_\omega \left[\sup_{\theta \in T} \langle \theta, \omega \rangle\right],$$

where $\omega = (\omega_1, \ldots, \omega_d)$ with $\omega_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$. Here, the inner product is the standard Euclidean product:

$$\langle \theta, \omega \rangle = \sum_{j=1}^{d} \theta_j \omega_j.$$

Definition 6 (Rademacher Process)**.** *The* Rademacher process *associated with $T \subset \mathbb{R}^d$ is*

$$\mathcal{R}(T) = \mathbb{E}_\varepsilon \left[\sup_{\theta \in T} \langle \theta, \varepsilon \rangle\right],$$

where $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_d)$ with $\varepsilon_i \overset{\text{i.i.d.}}{\sim} \text{Rad}(1/2)$.

Example. For the ℓ_2-ball B_2^d, $\mathcal{R}(B_2^d) = \mathbb{E}_\varepsilon\left[\sup_{\theta \in B_2^d} \langle \theta, \varepsilon \rangle\right]$. By the Cauchy–Schwarz inequality,

$$|\langle \theta, \varepsilon \rangle| \le \|\theta\|_2 \|\varepsilon\|_2 \le \|\varepsilon\|_2 \le \sqrt{d}$$

We can write

$$\langle \theta, \varepsilon \rangle = \sum_{i=1}^{d} \theta_i \varepsilon_i$$

The supremum is attained when

$$\theta_i^* = \frac{\operatorname{sgn}(\varepsilon_i)}{\sqrt{d}}, \qquad \|\theta^*\|_2 = 1$$

Then

$$\sum_{i=1}^{d} \theta_i^* \varepsilon_i = \frac{1}{\sqrt{d}} \sum_{i=1}^{d} \varepsilon_i \operatorname{sgn}(\varepsilon_i) = \sqrt{d}$$

Hence,

$$\mathcal{R}(B_2^d) = \sqrt{d}$$

Similarly, for Gaussian width $\mathbb{E}\Big[\sup_{\theta \in B_2^d} \langle \theta, \omega \rangle\Big]$. By the Cauchy–Schwarz inequality,

$$\langle \theta, \omega \rangle \le \|\theta\|_2 \|\omega\|_2 \le \|\omega\|_2$$

and equality is attained at $\theta = \omega / \|\omega\|_2$. Thus,

$$\mathcal{G}(B_2^d) := \mathbb{E}\left[\sup_{\theta \in B_2^d} \langle \theta, \omega \rangle\right] = \mathbb{E}\|\omega\|_2 = \mathbb{E}\sqrt{\|\omega\|_2^2} \le \sqrt{\mathbb{E}\|\omega\|_2^2} = \sqrt{d}$$

The inequality is obtained through Jensen's inequality, And we leave as an exercise to prove:

$$\mathbb{E}\|\omega\|_2 \ge \sqrt{d}\,(1 - o_d(1))$$

So,

$$\mathcal{G}(B_2^d) = \sqrt{d}(1 - o_d(1)) \qquad \mathcal{R}(B_2^d) = \sqrt{d}$$

1.3 Gaussian Process

Definition 7 (Gaussian Process)**.** *A stochastic process $\{U_\theta\}_{\theta \in \Theta}$ is **Gaussian** if, for every finite collection $\{\theta_1, \ldots, \theta_N\} \subset \Theta$, the random vector $(U_{\theta_1}, \ldots, U_{\theta_N})$ is jointly Gaussian with mean zero.*

Intuition: Let $U_\theta = \langle \theta, \omega \rangle, \omega \sim \mathcal{N}(0, I_d)$. Then $\{U_\theta\}_{\theta \in \mathbb{R}^d}$ is a Gaussian process. For any $\theta, \theta' \in \mathbb{R}^d$,

$$U_\theta - U_{\theta'} = \langle \theta - \theta', \omega \rangle \sim \mathcal{N}\big(0, \|\theta - \theta'\|_2^2\big)$$

Therefore, for any $\lambda \in \mathbb{R}$, we can connect gaussian process with metric:

$$\mathbb{E}\Big[e^{\lambda(U_\theta - U_{\theta'})}\Big] = \exp\left(\frac{\lambda^2 \|\theta - \theta'\|_2^2}{2}\right) = \exp\left(\frac{\lambda^2 d(\theta, \theta')^2}{2}\right)$$